

Financial Econometrics: Return Predictability I

Sora chon

Inha University

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

Outline

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

Unconditional and conditional expectation

- ▶ What is the average (or expected) temperature in South Korea?

$$\mathbb{E}(Y) = 13.3$$

- ▶ What is the average temperature in summer in South Korea?

$$\mathbb{E}(Y|\text{summer}) = 24.5$$

Unconditional expectation: $\mathbb{E}(Y) = \int y p(y) dy$

Conditional expectation: $\mathbb{E}(Y|X = x) = \int y p(y|X = x) dy$

where $p(\cdot)$ is the probability density function.

Law of iterative expectation

- ▶ Consider two random variables X, Y .
- ▶ Formally: The unconditional expectation of a random variable is equal to the expectation of the conditional expectation of the random variable conditional on some other random variable

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}[Y \mid X])$$

- ▶ Informally: the weighted average of the conditional averages is the unconditional average

LIE: Proof for the discrete case

$$\begin{aligned}\mathbb{E}(\mathbb{E}[Y \mid X]) &= \sum_x \mathbb{E}[Y \mid X = x] p(x) \\&= \sum_x \left(\sum_y y p(y \mid x) \right) p(x) \\&= \sum_x \sum_y y p(x, y) \\&= \sum_y y \sum_x p(x, y) \\&= \sum_y y p(y) \\&= \mathbb{E}(Y)\end{aligned}$$

LIE: Proof for the continuous case

$$\begin{aligned}\mathbb{E}[\mathbb{E}(Y | X)] &= \int \mathbb{E}(Y | X = u) g_x(u) du \\&= \int \left[\int t f_{y|x}(t | X = u) dt \right] g_x(u) du \\&= \iint t f_{y|x}(t | X = u) g_x(u) du dt \\&= \int t \left[\int f_{y|x}(t | X = u) g_x(u) du \right] dt \\&= \int t [f_{x,y} du] dt \\&= \int t g_y(t) dt \\&= \mathbb{E}(y)\end{aligned}$$

To avoid confusion, I use u as a value of X and use t as a value of Y .

Outline

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

Motivation

- ▶ Can we predict future stock returns ahead of time?
- ▶ Why do stock prices vary a lot?

Robert J. Shiller



Figure: Robert J. Shiller

On October 14, 2013, it was announced that Shiller had received the 2013 Nobel Prize in Economics alongside Eugene Fama and Lars Peter Hansen.

Robert J. Shiller

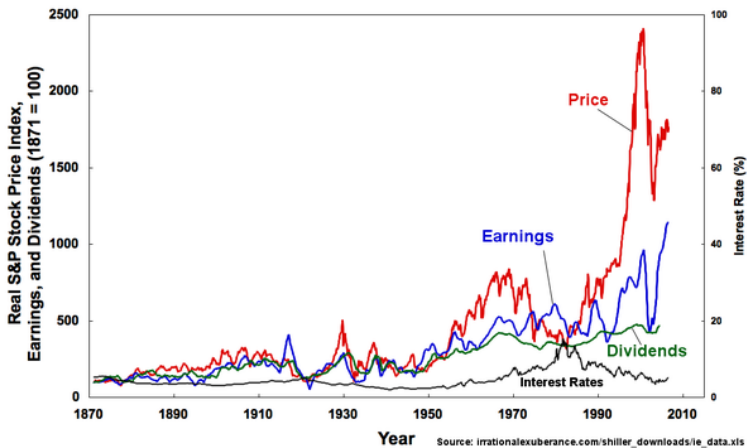


Figure: Robert J. Shiller

Robert J. Shiller

- ▶ Shiller argued that in a rational stock market, investors would base stock prices on the expected receipt of future dividends, discounted to a present value.
- ▶ He examined the performance of the U.S. stock market since the 1920s, and considered the kinds of expectations of future dividends and discount rates that could justify the wide range of variation experienced in the stock market.
- ▶ Shiller concluded that the volatility of the stock market was greater than could plausibly be explained by any rational view of the future.
- ▶ Reference: Shiller, Robert J. "Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?." The American Economic Review 71.3 (1981): 421-436.

Motivation: Traditional view ($\sim$ 80s)

- ▶ Expected returns don't move much over time - stocks are unpredictable beyond the unconditional means.
- ▶ Stock prices move on the news of cashflows (dividends).

Motivation: Current view

- ▶ Expected returns move a lot over time - stocks are predictable. (It moves with business cycle.)
- ▶ Stock prices move on the news of discount rate (expected return).

Outline

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

OLS evidence for traditional view

Consider the following simple linear regression model

$$R_{t+1} = a + bx_t + \varepsilon_{t+1}$$

where R_{t+1} is the gross return at time $t + 1$, x_t is a predictor (signal) that we can observe at time t , and ε_{t+1} is the prediction error at time $t + 1$.

- ▶ What is the conditional expectation of return R_{t+1} given x_t ?
 $E_t[R_{t+1}] = a + bx_t$ because $E_t[\varepsilon_{t+1}] = 0$. (note: $E_t[\cdot]$ is the conditional expectation given information set available at time t . The prediction error is something we cannot predict!)
- ▶ The information set at time t includes the predictor x_t .
- ▶ If $b = 0$, $E_t[R_{t+1}] = a$. The expected return does not vary. A constant expected return means that the return is not predictable beyond the unconditional mean a .
- ▶ If $b \neq 0$, $E_t[R_{t+1}] = a + bx_t$. The expected return varies. A varying expected return means that the return is predictable.

OLS evidence for traditional view

	$R_{t+1} = a + bR_t + \varepsilon_{t+1}$				
	b	t(b)	R ²	E(R)	$\sigma(E_t(R_{t+1}))$
Stock	0.04	0.33	0.002	11.4	0.77
T bill	0.91	19.5	0.83	4.1	3.12
Excess	0.04	0.39	0.00	7.25	0.91

- ▶ The above result is obtained using the annual data from 1927 to 2008.
- ▶ 1965-1985 view: Expected returns are constant over time.
- ▶ These regressions are usually higher frequency. Note R^2 and $\sigma(E(R))$ typically are even smaller at high frequency.
- ▶ This type of the regression evidence supports the basic "efficient markets" logic which stems from competition.

OLS evidence for traditional view

- ▶ "Efficient market" hypothesis
- ▶ Information is disseminated instantly + severe competition
- ▶ A current asset price contains all available information on the asset. And because of that, it is not possible to predict the price in the future.
- ▶ e.g.

OLS evidence for traditional view

	$R_{t+1} = a + bR_t + \varepsilon_{t+1}$				
	b	t(b)	R ²	E(R)	$\sigma(E_t(R_{t+1}))$
Stock	0.04	0.33	0.002	11.4	0.77
T bill	0.91	19.5	0.83	4.1	3.12
Excess	0.04	0.39	0.00	7.25	0.91

- ▶ Excess return: $R_{t+1} - \text{T bill rate (risk-free rate)}$
- ▶ The excess return represents the compensation of bearing risk in the financial market.
- ▶ The gross return represents both the risk compensation and the time compensation.

OLS evidence for traditional view

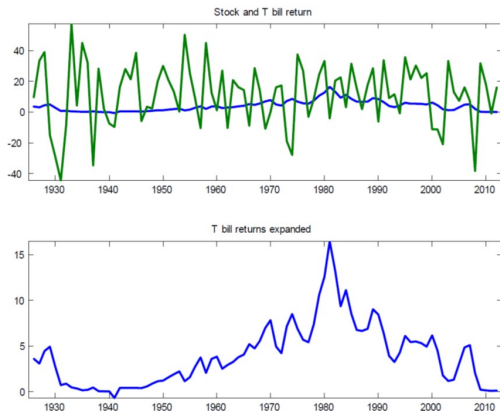


Figure 3: Time series of stock and T bill returns.

- In practice, it does not matter which one (gross or excess return) to use. Empirical results are very similar.

Outline

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

Dividend-price ratio (or Price-dividend ratio)

Intuition: a higher return comes from a higher price in the future, a lower initial price, or a higher dividend.

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

or

Intuition: The current price is the present value of future cash flows discounted by the return.

$$P_t = \frac{P_{t+1} + D_{t+1}}{R_{t+1}}$$

Dividend-price ratio

- By repeatedly substituting future prices, we have

$$P_t = \frac{D_{t+1}}{R_{t+1}} + \frac{D_{t+2}}{R_{t+1}R_{t+2}} + \frac{D_{t+3}}{R_{t+1}R_{t+2}R_{t+3}} + \dots$$

- The current price that is driven by future expectations is the sum of infinitely many future dividends discounted by returns.

$$P_t = \sum_{k=1}^{\infty} E_t \left(\frac{D_{t+k}}{R_{t+1}R_{t+2}R_{t+3}..R_{t+k}} \right)$$

where $E_t[.]$ is the conditional expectation operator.

- That is, the current price is the present value of all future dividends.

Dividend-price ratio

- ▶ Ok... We get it.. but let's further simplify it to get to more intuition on the information content of the dividend-price ratio.
- ▶ Assume one period case.

$$R_{t+1} = \frac{D_{t+1}}{P_t}, \quad E_t(R_{t+1}) = \frac{E_t(D_{t+1})}{P_t}$$
$$P_t = \frac{E_t(D_{t+1})}{E_t(R_{t+1})}, \quad \frac{P_t}{D_t} = \frac{E_t(D_{t+1}/D_t)}{E_t(R_{t+1})}$$

where $E_t[.]$ is the conditional expectation operator.

- ▶ Log are nicer so we can use linear time series methods,

$$r_{t+1} = d_{t+1} - p_t;$$
$$p_t - d_t = \Delta d_{t+1} - r_{t+1}$$
$$p_t - d_t = E_t(\Delta d_{t+1}) - E_t(r_{t+1}).$$

Dividend-price ratio

$$p_t - d_t = E_t(\Delta d_{t+1}) - E_t(r_{t+1})$$

- ▶ This is the key equation for the study of the return predictability.
- ▶ When traders obtain new information about future dividends (Δd_{t+1}), they form $E_t(\Delta d_{t+1})$. Then, the current (log) price responds.
- ▶ When traders obtain new information about future returns (r_{t+1}), they form $E_t(r_{t+1})$. Then, the current (log) price responds.
- ▶ Although we cannot see the information that the traders see, we can see the current price (P_t) along with the current dividend (D_t).

Dividend-price ratio

$$p_t - d_t = E_t(\Delta d_{t+1}) - E_t(r_{t+1})$$

- ▶ The current price contains the information about market participants' expectations for the future!
- ▶ Question: What derives the fluctuation of p_t ? Is it the news on dividend growth? or Is it the news on the discount rate (or expected return)?

Dividend-price ratio

$$p_t - d_t = E_t(\Delta d_{t+1}) - E_t(r_{t+1})$$

- ▶ Think about what influence the dividend growth and the discount rate.
- ▶ The dividend growth is typically determined by firms growth potentials and dividend payout policies.
- ▶ The discount rate (expected return) is typically determined by the overall amount of various risk factors.

Outline

Preliminary: Expectation

Motivation

Traditional view for return predictability

Information on D/P ratio

New view for return predictability

OLS evidence for new view

Horizon k	$R_{t \rightarrow t+k}^e = a + b \frac{D_t}{P_t} + \varepsilon_{t+k}$			
	b	$t(b)$	R^2	$\sigma [E_t(R^e)]$
1 year	3.8	(2.6)	0.09	5.46

Table: Return forecasting regressions $R_{t \rightarrow t+k}^e = a + b \frac{D_t}{P_t} + \varepsilon_{t+k}$;
Return data from 1947 to 2009.

- ▶ $R_{t \rightarrow t+k}^e$ is the excess return of holding the market portfolio from time t to time $t+k$.
- ▶ D_t is the current dividend. P_t is the current price. $\frac{D_t}{P_t}$ is referred to as the dividend-price ratio or dividend yield.
- ▶ Focus on the economic size of b . It is huge! 10% R^2 is also large considering that the stock return is volatile.

OLS evidence for new view

OLS evidence for new view

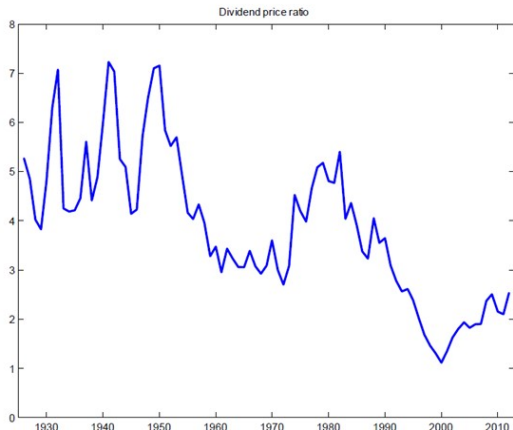


Figure: Dividend-price ratio

- How much does the expected return vary? It is huge!

OLS evidence for new view

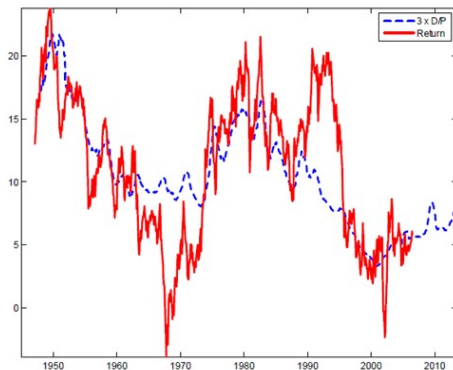


Figure: Dividend yield (multiplied by 3) and following 7 year return. CRSP VW market index.

- The expected return and realized return matches pretty well!

OLS evidence for new view



Figure: the log dividend-price ratio for the CRSP value weighted index and the log earnings-price ratio (10-year moving average) for the S&P 500. 1926:4-2002:4

- We use D/P only for simplicity. There are lots of variables that forecast returns. In fact, the main area of research focus right now is to explore all of these to build reliable forecasting models.

OLS evidence for new view

- ▶ Now, we have evidence that the price dividend ratio has a predictive power for the stock market return.
- ▶ The unsolved question is... what is the driving force for the variations in the dividend-price ratio? Discount rate (expected return) or dividend growth?

Dividend-price ratio

Horizon k (years)	$R_{t \rightarrow t+k}^e = a + b \frac{D_t}{P_t} + \varepsilon_{t+k}$			$\frac{D_{t+k}}{D_t} = a + b \frac{D_t}{P_t} + \varepsilon_{t+k}$		
	b	t(b)	R ²	b	t(b)	R ²
1	4.0	2.7	0.08	0.07	0.06	0.0001
2	7.9	3.0	0.12	-0.42	-0.22	0.001
3	12.6	3.0	0.20	0.16	0.13	0.0001
5	20.6	2.6	0.22	2.42	1.11	0.02

Table. OLS regressions of excess returns (value weighted NYSE - treasury bill) and real dividend growth on the value weighted NYSE dividend-price ratio.

Sample 1927-2005, annual data. $R_{t \rightarrow t+k}^e$ denotes the total excess return from time t to time $t+k$. Standard errors use GMM (Hansen-Hodrick) to correct for heteroskedasticity and serial correlation.

- ▶ **Once again with the different dataset, the market return is predictable. What it means by the predictability is that the dividend price ratio reflects news on the future return. And the market's expectation has an actual predictive power.**
- ▶ **The dividend growth is not predictable. What it means is that the dividend price ratio does not reflect news on the dividend growth.**

Dividend-price ratio

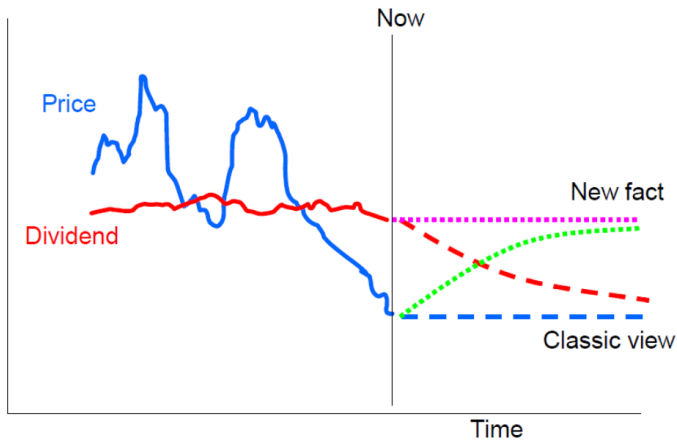


Figure: Comparison between new and old views

Dividend-price ratio

- ▶ When the price is low relatively to the dividend.. high dividend-price ratio..
- ▶ Old view: The future return is not predictable because the market is efficient. The current price is the best predicted price! The low price must be associated with low dividends in the future! ($\frac{D_{t+1}}{D_t}$ is expected to fall..)
- ▶ New view: The future return is predictable but the dividend growth is not predictable. The low price must be associated with high return in the future! The price will go up in the future!

Main takeaways

- ▶ 1: The market price (relative to D_t) varies a lot over time.
- ▶ 2: To explain the huge variations in the market price, it must be the case that $E_t(R_{t+1})$ changes or $E_t\left(\frac{D_{t+1}}{D_t}\right)$ changes or both.

More economic stories

- ▶ Traders has new information that a financial crisis will come in the near future. For the reason, they sell stocks or don't buy stocks as much as before. The current stock price significantly falls (relative to dividends). Buying a good stock with a very low price becomes a good deal because it brings a high expected return. A low price (high dividend-price ratio) is associated with a high expected return.
- ▶ Traders has new information that firms pay more dividends in the future. This news will encourage traders to buy stocks in the stock market. Consequently, stock prices will go up. A high price (low dividend-price ratio) is associated with a high expected dividend growth.
- ▶ But our empirical evidence tells us that although the story about dividends is appealing, it is not the main reason for the stock market movements!

Several weeks later, we will be talking about other predictors and other prediction models (based on the present value approach)!